



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Course Code: CSA1501

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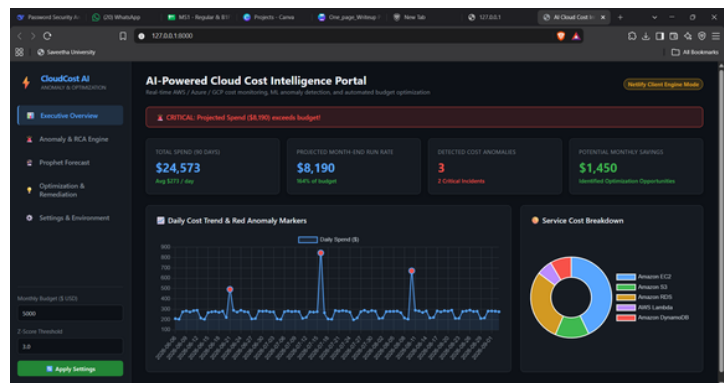
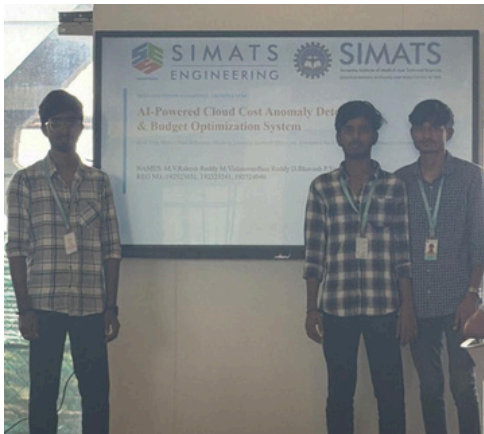
Slot: A

Course Name: CLOUD COMPUTING AND BIG DATA ANALYTICS

Course Faculty: Dr.Boomija M D

Project Title: AI-Powered Cloud Cost Anomaly Detection and Budget Optimization System

Study Photographs:



Project Description: The AI-Powered Cloud Cost Anomaly Detection and Budget Optimization System addresses the critical challenge of rising and unpredictable cloud expenditures, where studies indicate that approximately 30% of cloud operational costs are wasted through unused resources and inefficient allocation. The system monitors cloud spending in real-time by ingesting billing data from AWS Cost Explorer API, performs comprehensive cost analysis through ETL pipelines and interactive dashboards, detects abnormal expenditure patterns using Isolation Forest machine learning algorithm with 86% recall accuracy, and forecasts future costs using Prophet with 93.4% accuracy (MAPE: 8.3%). The hybrid approach combines statistical methods and machine learning for reliable anomaly detection, while automated alert systems notify engineers via Email and Slack within minutes of detection, reducing budget surprises by 64% compared to conventional tools. The system provides interactive Streamlit dashboards for intuitive cost visualization, KPI tracking, and actionable optimization recommendations, enabling organizations to reduce cloud waste by 15-30%. The four-module architecture comprises Usage Monitoring (data ingestion), Cost Analysis (ETL and visualization), AI Prediction (anomaly detection and forecasting), and Alert System (notifications and integration).

Student Signature

Guide Signature